

Deep Learning Project

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Latent Space Representation of Coding Experience

Introduction

The objective of this project is to analyze student coding proficiency using embedding techniques. By representing coding experience as a vector in $\mathbb{R}^6$, we can apply linear classification models to identify students who have undergone formal programming training versus those who have not.

Methodology

- Dataset Preparation

A synthetic dataset of 1000 student entries was generated, where each entry $x \in \mathbb{N}^6$ represents the number of lines of code written in C/C++, Python, MATLAB, R, Assembly, and Java.

- Experience Scoring

A custom function was designed to calculate the total coding experience E for a student. This scalar value provides a foundational metric for assessing overall proficiency.

- Perceptron Model

We implemented a single-layer perceptron to solve the binary classification problem:

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* **Feature Space:**  $\mathcal{X} = \mathbb{N}^6$   
* **Outcome Space:**  $\mathcal{Y} = \{ \text{Has taken}, \text{Hasn't taken} \}$   
* **Decision Rule:**  $w^T x \geq b$ 
```

The algorithm iteratively updates the weight vector w and bias b based on classification errors until the data is partitioned correctly.

Visualization and Analysis

To visualize the model's decision-making, we restricted the feature space to two dimensions: C/C++ experience and Python experience.

The red line in the plot represents the boundary $w^T x = b$. Students positioned above this boundary are classified as having taken a programming class, while those below are classified as not having taken one.

Results

The perceptron model successfully partitioned the 1000-student dataset into two distinct groups. The convergence of the model demonstrates that coding experience in specific languages, when weighted correctly, acts as a reliable predictor for formal programming background.

Conclusion

This project demonstrates the effectiveness of the Perceptron algorithm in binary classification tasks. By mapping student experience into a latent feature space, we can interpret complex coding behaviors as simple linear partitions.